

Willie Rush Lim

Employment

Brown University, RI	2026 – current	Dean's Tamarkin Assistant Professor
	2024 – 2026	Tamarkin Assistant Professor

Education

2019 – 2024	PhD in Mathematics, Stony Brook University, NY ○ Dissertation: "<i>From Herman Rings to Herman Curves</i>" ○ Advisor: Dzmitry Dudko
2015 – 2019	MSci. in Mathematics, Imperial College London, UK

Research Interests

Complex dynamics and renormalization theory, at the interface of analysis and geometry

Publications

- [3] Hyperbolicity of renormalization of critical quasicircle maps
[Comm. Math. Phys.](#) 407 (2026): 88. (94 pages)
- [2] Rigidity of J-rotational rational maps and critical quasicircle maps
[Trans. Amer. Math. Soc.](#) 379 (2026): 2507–2567.
- [1] A priori bounds and degeneration of Herman rings with bounded type rotation number
[Invent. Math.](#) 242 (2025): 827–893.

Preprints

- [7] Rigidity of the attractor of neutral renormalization (with D. Dudko and M. Lyubich)
- [6] Lebesgue measure of the postcritical set of neutral quadratic polynomials
- [5] Uniform bounds for bubbles of neutral quadratic polynomials. (with D. Dudko) [arxiv:2609.15500](https://arxiv.org/abs/2609.15500)
- [4] The combinatorics of sector renormalization. [arXiv:2607.11408](https://arxiv.org/abs/2607.11408)

Research Visits

2026	July	Isaac Newton Institute for Mathematical Sciences, Cambridge, UK Invited participant for the Complex Dynamics: Interactions and Influences programme (4 weeks)
2026	April – May	Mathematical Institute of the Polish Academy of Sciences, Warsaw, Poland Invited early career researcher position for the Simons Semester: Continued Fractions, Fractals, Ergodic theory and Dynamics (5 weeks), gave a mini-course totalling 4.5 hours

Invited Talks

2026	July	Complex dynamics: a century on from Fatou and Julia, Isaac Newton Institute, UK
2026	June	Dynamical Systems Seminar, Stony Brook University
2026	April	Geometry and Topology Seminar at Brown and Yale (GATSBY)
2026	April	Arithmetic, Algebraic, and Complex Dynamics Seminar, Brown University
2025	August	Fields-CNAM Nonlinear Days 2025, Fields Institute, Canada
2025	August	Geometric Complexity of Julia Sets VI, Bedlewo, Poland
2025	June	Topics in Complex Dynamics, University of Barcelona, Spain
2025	February	Geometry & Topology Seminar, Brown University
2024	November	Dynamical Systems Seminar, Stony Brook University
2024	November	Dynamical Systems Seminar, Cornell University
2024	October	Dynamical Systems Seminar, Boston University
2024	April	AMS Sectional Meeting, University of Wisconsin–Milwaukee
2024	January	Geometry & Topology Seminar, Brown University
2023	March	Complex Analysis and Dynamics Seminar, City University of New York
2022	November	Dynamical Systems Seminar, Stony Brook University

Teaching Experience

At Brown University:

2026	Fall	Instructor, MATH1710 Topology
2026	Fall	Instructor, MATH1110 Ordinary Differential Equations
2026	Spring	Instructor, MATH1040 Fundamental Problems of Geometry
2025	Fall	Instructor, MATH1710 Topology
2025	Fall	Instructor, MATH1110 Ordinary Differential Equations
2025	Spring	Lecturer, MATH180 Multivariable Calculus
2024	Fall	Lecturer, MATH200 Multivariable Calculus

At Stony Brook University:

2023	Summer	Instructor, MAT127 Calculus C (part of Simons STEM Scholars Summer Bridge)
2022	Fall	Lecturer, MAT125 Calculus A
2022	Summer	Instructor, MAT203 Calculus III
2021	Summer	Instructor, MAT203 Calculus III
2020	Summer	Instructor, MAT342 Applied Complex Analysis
2019 – 2024		Teaching Assistant (as a grader and recitation leader) Calculus I, II, IV, Introduction to Linear Algebra, Applied Real Analysis, Applied Complex Analysis, Dynamics and Chaos, Graduate Complex Analysis

Honors and Awards

2026	Dean's Tamarkin Assistant Professorship, Brown University (department-wide research honor, one recipient)
2024	Chairman's Award for Excellence in Teaching, Stony Brook University
2019	IBM Prize for Excellence in Pure Mathematics
2019	Prize for Excellence in Support of Teaching, Imperial College London
2017	G-Research Prize
2015 – 2019	Dean's List, Imperial College London
2015 – 2019	President's Undergraduate Scholarship, Imperial College London

Service and Outreach

2026 – present	Reviewer for Mathematical Reviews (MathSciNet)
2022 Fall	Co-organizer of ENYGMMa (Empowering New York Gender Minority Mathematicians)
2022 Summer	Instructor of the Mathematics Summer Program (MathCamp) for high school students
2021 – 2022	Directed Reading Program
2020 – 2021	Co-organizer of the Graduate Student Seminar at Stony Brook University